

Scalable Multi-Agent Reinforcement Learning for Interference-Aware Spectrum Planning in Large-Scale UAV Swarms

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Abstract—The rapid proliferation of unmanned aerial vehicle (UAV) swarms in surveillance, relay, and delivery applications introduces unprecedented challenges in spectrum management. As swarm sizes scale to tens or hundreds of UAVs, mutual interference among co-located communication links becomes the dominant factor limiting system reliability. Traditional centralized spectrum allocation methods suffer from combinatorial explosion, while independent optimization strategies fail to account for inter-agent coupling, leading to suboptimal frequency reuse. In this paper, we propose a multi-agent deep reinforcement learning (MADRL) framework based on the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm to jointly optimize power and frequency allocation in large-scale UAV swarms. Our approach leverages the centralized training with decentralized execution (CTDE) paradigm, where a centralized critic is trained with global state information while each UAV executes decisions using only local observations of itself and its nearest neighbors. We introduce a Gumbel-Softmax discrete action parameterization that enables gradient-based optimization while respecting practical discrete channel and power-level constraints. A sequential decision protocol with rotating order further aligns the method with real-world asynchronous deployment scenarios. Comprehensive simulations across swarm sizes ranging from 10 to 100 UAVs demonstrate that the proposed approach significantly outperforms greedy baselines and independent RL methods, achieving lower interference probability while maintaining scalability. Ablation studies reveal the importance of local observation design, hybrid reward structures, and temperature annealing in enabling effective coordination at scale.

Index Terms—UAV swarm, spectrum planning, multi-agent reinforcement learning, MADDPG, Gumbel-Softmax, interference management

I. INTRODUCTION

A. Motivation

Unmanned aerial vehicle (UAV) swarms are emerging as a transformative technology for a wide range of civil and military applications, including disaster surveillance, environmental monitoring, communication relay, and package delivery [1], [2]. A defining characteristic of future UAV swarm operations is the need for reliable intra-swarm communication links operating in shared spectrum bands. As the number of co-deployed UAVs grows from tens to hundreds, the electromagnetic spectrum becomes a critically scarce resource, and mutual interference among nearby communication transceivers rapidly degrades link reliability [3], [4].

The frequency planning problem for UAV swarms is fundamentally different from classical cellular spectrum allocation

for several reasons. First, UAV swarms operate in three-dimensional space with highly dynamic topologies. Second, communication links are typically formed through ad-hoc pairing among nearby UAVs rather than through fixed base-station associations. Third, the interference structure is inherently local: a UAV transmission primarily affects receivers within a few hundred meters, creating natural spatial reuse opportunities that must be exploited efficiently [5].

B. Problem Statement

Consider a swarm of N UAVs deployed in a 2000×2000 m operational area. Each UAV carries one transmitter-receiver pair operating with a bandwidth of 10 MHz within the 1200–1300 MHz frequency band. The transmit power of each UAV can be continuously adjusted between 0 and 5 W. Communication links are formed by pairing each transmitter with a receiver of a different UAV within a 500 m effective range. At each receiver, the Signal-to-Interference-plus-Noise Ratio (SINR) depends on the desired signal power, the aggregate interference from all other co-channel or partially overlapping transmissions, and the thermal noise floor. A receiver is considered to experience interference when its SINR falls below a predefined threshold. The *interference probability*—the fraction of receivers with sub-threshold SINR—serves as the primary performance metric.

The optimization problem is to jointly select the center frequency and transmit power for each UAV such that the interference probability is minimized, subject to per-link SINR requirements. Formally, let $f_i \in [f_{\min}, f_{\max}]$ and $p_i \in [0, P_{\max}]$ denote the frequency and power of UAV i , respectively. The objective is:

$$\min_{\{f_i, p_i\}_{i=1}^N} \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{SINR}_i(f_1, \dots, f_N, p_1, \dots, p_N) < \gamma_{\text{th}}], \quad (1)$$

where $\mathbb{I}[\cdot]$ is the indicator function and γ_{th} is the SINR threshold.

C. Limitations of Existing Approaches

Traditional approaches to this problem can be categorized into three groups. *Graph-theoretic methods* model interference as a conflict graph and employ graph coloring algorithms to assign orthogonal channels to interfering links [6], [7].

While computationally tractable, these methods rely on binary interference models that ignore the continuous nature of SINR degradation and power adjustments. *Centralized optimization* approaches, such as mixed-integer programming or exhaustive search, can produce near-optimal solutions for small swarms but become computationally prohibitive as N grows—the joint action space of power-frequency combinations grows combinatorially with the number of UAVs [8]. *Greedy heuristic methods* make locally optimal decisions for one UAV at a time but are sensitive to the decision order and cannot recover from suboptimal early choices, particularly when the swarm density is high [9].

In the reinforcement learning (RL) domain, *independent RL* approaches—where each UAV learns its own policy treating other UAVs as part of the environment—have been applied to spectrum access problems [12], [13]. However, the non-stationarity introduced by concurrent learning agents leads to unstable training and credit assignment failures, especially as the number of agents increases. *Centralized RL*, where a single policy outputs joint actions for all UAVs, faces an exponentially growing action space and does not scale beyond small swarms [22].

D. Why MARL and Our Approach

Multi-agent reinforcement learning (MARL) offers a principled framework to address these limitations through the centralized training with decentralized execution (CTDE) paradigm [18], [23]. In CTDE, agents are trained with access to global information (centralized critic), but during execution, each agent makes decisions using only local observations (decentralized actor). This architecture is particularly well-suited to the UAV frequency planning problem because the interference structure is intrinsically local: a UAV’s transmission primarily affects receivers within its spatial neighborhood. By restricting each UAV’s observation to its own state and the states of its K nearest neighbors, we achieve a scalable representation that captures the relevant interference couplings without the dimensionality explosion of full global observations.

In this paper, we propose a MADDPG-based framework with three key innovations:

- 1) **Gumbel-Softmax Discrete Action Parameterization:** Rather than treating frequency and power as continuous outputs, we discretize them into practical levels and employ the Gumbel-Softmax reparameterization [27] to enable differentiable sampling from categorical distributions. This aligns with real-world hardware constraints (discrete channel grids, finite power-step resolution) while preserving end-to-end gradient-based training.
- 2) **Sequential Decision Protocol:** We introduce a rotating-order sequential decision scheme where UAVs commit to their frequency-power selections one at a time, with later agents observing the decisions of earlier ones. This protocol reflects practical asynchronous deployment scenarios and provides a natural mechanism for coordination without explicit communication.

- 3) **Hybrid Reward Design:** Our reward function combines a global interference penalty (shared by all agents to promote cooperation) with per-agent terms for link quality, threshold satisfaction, power efficiency, and local congestion awareness. This hybrid structure prevents the free-rider problem while encouraging globally optimal behavior.

The main contributions of this paper are:

- We formulate the large-scale UAV swarm frequency planning problem as a multi-agent Markov decision process with a local-observation CTDE architecture that naturally exploits the spatial locality of interference.
- We develop a MADDPG-based framework with Gumbel-Softmax discrete action parameterization, sequential decision protocol, and hybrid reward design tailored for practical spectrum allocation.
- We conduct comprehensive experiments across swarm sizes from 10 to 100 UAVs, comparing against random, greedy (centralized, sequential, independent), independent DDPG (IDDPG), and centralized DDPG (CDDPG) baselines.
- Through ablation studies, we analyze the contributions of individual design components and provide insights into emergent frequency reuse patterns learned by the MARL agents.

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. Section III describes the proposed MARL-based frequency planning framework. Section IV presents simulation experiments and results. Section V discusses findings and limitations. Section VI reviews related work, and Section VII concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. UAV Swarm Communication Model

We consider a swarm of N UAVs operating in a two-dimensional area of size $L \times L$ ($L = 2000$ m). Each UAV i has a three-dimensional position $\mathbf{s}_i = (x_i, y_i, z_i)$, where z_i is uniformly distributed in $[50, 150]$ m representing hovering altitude variation. UAVs are deployed with uniformly random spatial distribution, yielding a node density of $\rho = N/(L/1000)^2$ UAVs per square kilometer.

Each UAV is equipped with one communication payload consisting of a transmitter and a receiver, both operating in half-duplex mode. Communication links are established through nearest-neighbor pairing: for each transmitter, we search for the closest receiver of a different UAV within an effective communication range $R_{\text{eff}} = 500$ m. The pairing process ensures that no two transmitters share the same receiver and no two receivers share the same transmitter, forming a bipartite matching between transmitters and receivers. Each link forms a unidirectional communication channel from transmitter to its paired receiver.

The radio parameters are configured as follows. Each link operates with a bandwidth $B = 10$ MHz within the frequency

band [1200, 1300] MHz, providing $n_{\text{ch}} = 11$ non-overlapping channels when using orthogonal frequency assignments. The transmit power is continuously adjustable: $p_i \in [0, P_{\text{max}}]$ with $P_{\text{max}} = 5$ W. Both transmitter and receiver antennas are assumed to be omnidirectional with 0 dBi gain for simplicity, though the model accommodates arbitrary antenna patterns. The thermal noise floor is $N_0 = -90$ dBm.

B. Interference Model

The received signal power at receiver j from transmitter i is determined by the free-space path loss model:

$$PL_{ij}^{\text{dB}} = 20 \log_{10}(d_{ij}^{\text{km}}) + 20 \log_{10}(f_i^{\text{MHz}}) + 32.44, \quad (2)$$

where d_{ij}^{km} is the Euclidean distance between transmitter i and receiver j in kilometers, and f_i^{MHz} is the transmission frequency in MHz.

When two links operate on different center frequencies, the interference coupling depends on the spectral overlap. Let $B_i = B$ denote the bandwidth of all links. Given transmitter i at center frequency f_i and receiver j tuned to frequency f_j , the fractional spectral overlap is:

$$\alpha_{ij} = \frac{\max(0, \min(f_i^+, f_j^+) - \max(f_i^-, f_j^-))}{B_i}, \quad (3)$$

where $f_i^{\pm} = f_i \pm B_i/2$ and $f_j^{\pm} = f_j \pm B_j/2$. The received interference power from transmitter i at receiver j is:

$$I_{ij}^{\text{dBm}} = 10 \log_{10}(p_i \cdot 1000 \cdot \alpha_{ij}) + G_t + G_r - PL_{ij}^{\text{dB}}. \quad (4)$$

The SINR at receiver j , which is paired with its desired transmitter $t(j)$, is:

$$\text{SINR}_j^{\text{dB}} = S_{t(j),j}^{\text{dBm}} - 10 \log_{10} \left(\sum_{i \neq t(j)} 10^{I_{ij}^{\text{dBm}}/10} + 10^{N_0/10} \right), \quad (5)$$

where $S_{t(j),j}^{\text{dBm}}$ is the desired signal power computed analogously to (4) with $\alpha_{t(j),j} = 1$.

The interference probability P_{int} is defined as the fraction of receivers with SINR below a threshold γ_{th} :

$$P_{\text{int}}(\mathbf{f}, \mathbf{p}) = \frac{1}{N} \sum_{j=1}^N \mathbb{I}[\text{SINR}_j^{\text{dB}} < \gamma_{\text{th}}]. \quad (6)$$

C. Optimization Problem

The frequency planning problem is formulated as:

$$\min_{\mathbf{f}, \mathbf{p}} P_{\text{int}}(\mathbf{f}, \mathbf{p}) \quad (7)$$

$$\text{s.t. } f_i \in [f_{\min} + B/2, f_{\max} - B/2], \quad \forall i \quad (8)$$

$$p_i \in [0, P_{\text{max}}], \quad \forall i \quad (9)$$

$$\text{SINR}_i \geq \gamma_{\text{th}}, \quad \forall i \quad (10)$$

where constraint (8) ensures the full signal bandwidth fits within the allocated band, and constraint (10) represents per-link quality-of-service requirements.

This problem is NP-hard in general. The joint optimization of $2N$ continuous variables with a non-convex, discontinuous

objective function (due to the indicator function in (6)) defies conventional gradient-based optimization. Furthermore, the interference coupling is non-additive: changing one UAV's frequency affects all receivers within its interference range, and the SINR at each receiver depends on the aggregate of multiple interferers. For a swarm of $N = 50$ UAVs with $n_{\text{ch}} = 11$ channels and $n_p = 10$ power levels, the joint action space size is $(n_{\text{ch}} \cdot n_p)^N \approx 110^{50}$, making exhaustive search infeasible.

D. Multi-Agent MDP Formulation

We formulate the frequency planning problem as a multi-agent Markov Decision Process (MDP) defined by the tuple $\langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where $\mathcal{N} = \{1, \dots, N\}$ is the set of UAV agents.

1) *Observation Space*: Each agent i receives a local observation $o_i \in \mathcal{O}$ constructed from its own state and the states of up to K nearest neighbors within an observation radius R_{obs} . The self-observation consists of:

- Normalized 3D position: $(x_i/L, y_i/L, (z_i - z_{\text{center}})/z_{\text{span}})$
- Normalized transmitter parameters: power (p_i/P_{max}), frequency $((f_i - f_{\min})/(f_{\max} - f_{\min}))$, SINR margin $(\tanh((\text{SINR}_i - \gamma_{\text{th}})/10))$
- Normalized receiver frequency

For each of the K nearest neighbors within R_{obs} , the observation appends: existence flag, relative 3D position, and the same normalized parameters as above. For the sequential decision variant, we additionally include the fraction of already-executed UAVs and per-neighbor commitment flags. With $K = 5$ neighbors, the observation dimension is $7 + 7K = 42$ (or 50 with sequential information), which remains constant regardless of the total swarm size N , ensuring scalability.

2) *Action Space*: Rather than using continuous outputs, we define discrete action spaces for power and frequency. The power levels are n_p uniformly spaced values in the normalized range $[-1, 1]$ mapping to $[0, P_{\text{max}}]$. The frequency levels are n_f uniformly spaced values mapping to $[f_{\min} + B/2, f_{\max} - B/2]$. During training, we sample from the joint categorical distribution over $n_p \times n_f$ discrete actions using the Gumbel-Softmax relaxation, which provides differentiable approximations of discrete samples. A temperature parameter τ controls the sharpness of the approximation: at high τ , the distribution is uniform (exploration); at low τ , it approaches a one-hot vector (exploitation).

3) *Reward Design*: The reward function combines global and local components:

$$r_i = r_{\text{global}} + r_i^{\text{link}} + r_i^{\text{threshold}} + r_i^{\text{power}} + r_i^{\text{congestion}} + r_i^{\text{high-SINR}}, \quad (11)$$

where:

$$r_{\text{global}} = -\lambda_g \cdot P_{\text{int}} \quad (\text{shared global penalty, } \lambda_g = 2000) \quad (12)$$

$$r_i^{\text{link}} = \lambda_l \cdot \tanh((\text{SINR}_i - \gamma_{\text{th}})/\sigma) \quad (\text{link quality, } \lambda_l = 30, \sigma = 5) \quad (13)$$

$$r_i^{\text{threshold}} = \begin{cases} +15, & \text{if } \text{SINR}_i \geq \gamma_{\text{th}} \\ -25, & \text{otherwise} \end{cases} \quad (14)$$

$$r_i^{\text{power}} = -\lambda_p \cdot (p_i/P_{\text{max}}) \quad (\text{power efficiency, } \lambda_p = 5) \quad (15)$$

$$r_i^{\text{congestion}} = -\lambda_c \cdot \min(1, n_i^{\text{int}}/3) \quad (\text{local congestion awareness, } \lambda_c = 3) \quad (16)$$

$$r_i^{\text{high-SINR}} = -\lambda_h \cdot \max(0, \text{SINR}_i - \gamma_{\text{th}} - \delta) \quad (\text{excessive SINR penalty, } \lambda_h = 30, \delta = 3) \quad (17)$$

The global term r_{global} aligns all agents toward the common objective of minimizing interference probability. The link quality and threshold terms ensure individual agents do not sacrifice their own link performance. The power penalty encourages energy-efficient operation. The congestion term provides local awareness of interference density. The high-SINR penalty prevents excessive power usage that could cause unnecessary interference to others without benefiting the transmitter's own link.

III. MARL-BASED FREQUENCY PLANNING FRAMEWORK

A. Centralized Training with Decentralized Execution

The CTDE architecture forms the backbone of our framework. During training, a centralized critic for each agent has access to the global state, defined as the concatenation of all agents' local observations, and the joint actions of all agents. This enables the critic to learn accurate value functions that account for all inter-agent interactions. During execution, each agent's actor network maps only its local observation to its own action, requiring no global information or inter-agent communication.

Formally, for agent i , the centralized action-value function is $Q_i(\mathbf{o}, \mathbf{a}; \theta_i^Q)$, where $\mathbf{o} = [o_1, \dots, o_N]$ is the concatenated global observation and $\mathbf{a} = [a_1, \dots, a_N]$ is the joint action. The deterministic policy is $\mu_i(o_i; \theta_i^\mu)$, parameterized by the actor network. The critic is trained to minimize the temporal-difference error:

$$\mathcal{L}(\theta_i^Q) = \mathbb{E}_{(\mathbf{o}, \mathbf{a}, r, \sigma')} \left[(Q_i(\mathbf{o}, \mathbf{a}) - y_i)^2 \right], \quad (18)$$

where $y_i = r_i + \gamma Q_i'(\mathbf{o}', \mathbf{a}')|_{a_j' = \mu_j'(o_j')}$ and Q_i', μ_j' are target networks. The actor is updated by maximizing the expected return:

$$\mathcal{L}(\theta_i^\mu) = -\mathbb{E}_{\mathbf{o}} [Q_i(\mathbf{o}, a_1, \dots, \mu_i(o_i), \dots, a_N)]. \quad (19)$$

Both actor and critic networks are implemented as three-layer multi-layer perceptrons (MLPs) with 128 hidden units each and ReLU activations. Target networks are updated via soft updates with $\tau = 0.01$.

B. Gumbel-Softmax Discrete Action Parameterization

A key design choice in our framework is the use of discrete action spaces parameterized by the Gumbel-Softmax distribution. While continuous action parameterizations (e.g., tanh-squashed Gaussian outputs) are common in actor-critic methods, they are poorly suited to the frequency planning problem for two reasons. First, practical radio hardware operates on discrete frequency channel grids and finite power-step resolutions; continuous outputs require post-hoc quantization that can lead to performance degradation. Second, the interference landscape is highly non-smooth with respect to frequency—small frequency shifts can abruptly change spectral overlap relationships, making gradient-based optimization through continuous outputs unstable.

The Gumbel-Softmax provides a continuous relaxation of the categorical distribution [27], [28]. Given logits $\mathbf{z} \in \mathbb{R}^d$ (from the actor network) and a temperature $\tau > 0$, the Gumbel-Softmax produces a probability vector $\mathbf{w} \in \Delta^{d-1}$ (the $(d-1)$ -simplex):

$$w_k = \frac{\exp((z_k + g_k)/\tau)}{\sum_{j=1}^d \exp((z_j + g_j)/\tau)}, \quad (20)$$

where $g_k \sim \text{Gumbel}(0, 1)$ are i.i.d. samples drawn as $g_k = -\log(-\log(u_k))$ with $u_k \sim \text{Uniform}(0, 1)$. The final continuous action is the weighted sum of the discrete level values:

$$a = \sum_{k=1}^d w_k \cdot v_k, \quad (21)$$

where v_k are the actual power or frequency level values. During evaluation, we use the hard (argmax) mode: $w_k = \mathbb{I}[k = \arg \max_j (z_j + g_j)]$ to select a single discrete action.

The actor network outputs separate logits for power and frequency, with dimensions n_p and n_f respectively. The temperature τ is annealed from an initial value $\tau_{\text{init}} = 1.5$ to a final value $\tau_{\text{final}} = 0.02$ during training according to an exponential decay schedule:

$$\tau(t) = \tau_{\text{final}} + (\tau_{\text{init}} - \tau_{\text{final}}) \cdot \exp(-\lambda_\tau \cdot \text{progress}), \quad (22)$$

where $\lambda_\tau = 5.0$ and $\text{progress} = (t - t_{\text{warmup}})/(T - t_{\text{warmup}})$ for $t \geq t_{\text{warmup}}$. During a warmup phase of t_{warmup} episodes, the temperature is held constant at τ_{init} to encourage exploration. This annealing scheme transitions the policy from broad exploration to fine exploitation.

C. Sequential Decision Protocol

In real-world UAV deployments, perfectly simultaneous decision-making is rarely feasible due to communication latency, processing delays, and coordination overhead. Motivated by this consideration, we introduce a sequential decision protocol with rotating order. In each episode, UAVs make decisions one at a time in a randomly initialized rotating order. Each UAV observes the decisions (commitment status) of previously executed UAVs in its neighborhood, enabling a natural form of implicit coordination.

The protocol works as follows. In each macro-step, we fix a starting offset k (rotated each step) and iterate through UAVs in order $k, k+1, \dots, N-1, 0, \dots, k-1$. For the m -th UAV in this order, the observation is augmented with:

- The fraction of already-executed UAVs: m/N
- For each observed neighbor: a commitment flag (whether the neighbor has already made its decision) and its normalized commit index

After making its decision, the UAV commits to its frequency and power selection. This commitment is reflected in the environment's SINR computation for subsequent decision-makers. The rotating order ensures fairness over the course of training, as each UAV experiences all possible positions in the decision sequence.

The sequential protocol offers several advantages. It eliminates the credit assignment ambiguity of simultaneous actions, as each agent's reward is computed in the context of previously committed decisions. It naturally breaks symmetries that can lead to convergence to suboptimal equilibria. It also provides a practical deployment path, as real swarms typically activate UAVs in a staggered manner.

D. Hybrid Reward Design and Ablation

The hybrid reward structure in (11) is designed to balance global cooperation with individual incentives. The global interference penalty ($\lambda_g = 2000$) dominates the reward magnitude, ensuring that all agents are primarily motivated to reduce overall interference. However, relying solely on the global term can lead to the free-rider problem, where some agents obtain good performance by exploiting the cooperation of others. The local terms address this by rewarding individual link performance and penalizing inefficient resource usage.

Each local term serves a specific purpose:

- r_i^{link} : Provides smooth reward shaping for SINR improvement, using the tanh function to saturate at both extremes and avoid unbounded gradients.
- $r_i^{\text{threshold}}$: Binary bonus/penalty for meeting the SINR threshold, creating a strong incentive to satisfy the QoS constraint.
- r_i^{power} : Penalizes high power usage, encouraging agents to find the minimum power that achieves the required SINR, which in turn reduces interference to others.
- $r_i^{\text{congestion}}$: Provides local awareness of how many interferers are affecting the agent's receiver, encouraging frequency diversity in dense regions.
- $r_i^{\text{high-SINR}}$: Penalizes excessive SINR margins (above $\gamma_{\text{th}} + 10$ dB), discouraging unnecessarily high transmission power that degrades global performance.

In our ablation studies (Section IV-E), we examine the contributions of individual reward components by selectively removing them and measuring the resulting performance degradation.

E. Training Algorithm

Algorithm III-E summarizes the complete training procedure. The training process begins with a warmup phase where

actions are sampled with high temperature to populate the replay buffer with diverse experiences. After warmup, the temperature is annealed, and network updates begin once the buffer contains sufficient samples.

[t] MADDPG Training for UAV Swarm Frequency Planning

Require: Swarm size N , episodes T , warmup episodes T_w , temperature schedule $\tau(t)$

Ensure: Trained policies $\mu_1, \dots, \mu_N$

- 1: Initialize replay buffer $\mathcal{D}$ with capacity $C = 150,000$
- 2: Initialize actor networks μ_i and critic networks Q_i for $i = 1, \dots, N$
- 3: Initialize target networks $\mu'_i \leftarrow \mu_i, Q'_i \leftarrow Q_i$
- 4: **for** episode = 1 to T **do**
- 5: Reset environment, get initial observations
- 6: Set temperature $\tau = \tau(\text{episode})$
- 7: Randomly select starting offset $k \sim \text{Uniform}(0, N-1)$
- 8: **for** step = 1 to max_steps **do**
- 9: Collect observations $\{o_i\}_{i=1}^N$
- 10: **if** sequential mode **then**
- 11: Execute UAVs in rotating order starting from k
- 12: Each UAV observes prior commitments
- 13: **else**
- 14: All UAVs select actions simultaneously
- 15: **end if**
- 16: Apply actions: $a_i \sim \text{GumbelSoftmax}(\mu_i(o_i), \tau)$
- 17: Step environment, get rewards $\{r_i\}$ and next observations
- 18: Store transition $(\mathbf{o}, \mathbf{a}, \mathbf{r}, \mathbf{o}')$ in $\mathcal{D}$
- 19: **if** episode $\geq T_w$ and $|\mathcal{D}| \geq B$ **then**
- 20: Sample minibatch of size $B = 192$
- 21: **for** each agent i **do**
- 22: Compute target: $y_i = r_i + \gamma Q'_i(\mathbf{o}', \mathbf{a}')$
- 23: Update critic by minimizing $\text{MSE}(Q_i(\mathbf{o}, \mathbf{a}), y_i)$
- 24: Update actor by maximizing $Q_i(\mathbf{o}, \mathbf{a})$ w.r.t. μ_i
- 25: Soft update targets: $\theta' \leftarrow \tau\theta + (1-\tau)\theta'$
- 26: **end for**
- 27: **end if**
- 28: Update observations
- 29: **end for**
- 30: **if** episode mod eval_interval = 0 **then**
- 31: Evaluate policy (no exploration, $\tau = \tau_{\text{final}}$)
- 32: **end if**
- 33: **end for**

IV. SIMULATION EXPERIMENTS

A. Experimental Setup

We implemented our framework in Python using PyTorch and conducted experiments on a workstation with an Intel Core i7 CPU. The simulation parameters are summarized in Table I. All experiments use a fixed random seed of 42 for reproducibility.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Operational area	2000×2000 m
UAV count N	10, 25, 50, 75, 100
Frequency band	1200–1300 MHz
Channel bandwidth	10 MHz
Number of channels	11
Transmit power range	0–5 W
Noise floor	−90 dBm
Path loss model	Free-space (Eq. 2)
SINR threshold γ_{th}	0 dB
Effective pairing range	500 m
Observation radius R_{obs}	600 m
Max observed neighbors K	5

TABLE II
MADDPG TRAINING HYPERPARAMETERS

Hyperparameter	Value
Actor learning rate	1×10^{-4}
Critic learning rate	1×10^{-3}
Discount factor γ	0.95
Soft update τ	0.01
Batch size	192
Replay buffer capacity	150,000
Hidden layer size	128
Initial temperature τ_{init}	1.5
Final temperature τ_{final}	0.05
Temperature decay λ_τ	5.0
Warmup episodes	50–80 (scale-dependent)
Update interval (steps)	5
Power levels n_p	10
Frequency levels n_f	20

1) *Baseline Methods*: We compare the proposed MADDPG method against seven baselines spanning three categories:

Heuristic baselines:

- **Random**: Each UAV uniformly randomly selects its power and frequency. Represents the worst-case performance without any planning.
- **Greedy-Central (Greedy-C)**: A centralized exhaustive search that enumerates all combinations of frequency and power for each UAV, selecting the combination that minimizes global interference probability. Equivalent to brute-force subset optimization.
- **Greedy-Sequential (Greedy-S)**: UAVs make decisions one at a time in a fixed order. Each UAV selects the power-frequency pair that minimizes its own SINR degradation considering already-selected decisions of prior UAVs. This is the traditional sequential greedy approach.
- **Greedy-Independent (Greedy-I)**: All UAVs simultaneously select the locally optimal power-frequency pair without considering the decisions of other UAVs.

RL baselines:

- **IDDPG (Independent DDPG)**: Each UAV trains an independent DDPG agent with its own local critic, treating other agents as part of the environment. No centralized information is shared.

- **CDDPG (Centralized DDPG)**: A single global DDPG agent outputs joint actions for all N UAVs. The action space dimension is $2N$.

MARL method (ours):

- **MADDPG**: Our proposed method with Gumbel-Softmax discrete actions, CTDE architecture, and hybrid reward.

2) *Evaluation Protocol*: Each method is evaluated on 50 independently generated random UAV layouts (episodes) at each swarm size. For RL methods, we report the best evaluation performance achieved during training. Performance is measured by the average interference probability across all episodes. For methods with randomness (random baseline, RL exploration), we report the mean and standard deviation over episodes.

B. Interference Probability Comparison

[Fig. 1: Interference probability vs. UAV count for all methods. X-axis: Number of UAVs (10, 25, 50, 75, 100). Y-axis: Interference probability (0–1). Multiple lines: Random (dashed black, near 1.0 at all scales), Greedy-I (orange), Greedy-S (green), Greedy-C (blue), IDDPG (red, high variance), CDDPG (purple, not scalable beyond 25), MADDPG (thick dark blue). Error bars show ± 1 standard deviation.]

[Table II: Interference probability comparison table. Columns: Method, 10 UAVs, 25 UAVs, 50 UAVs, 75 UAVs, 100 UAVs. Shows mean $\pm$ std for each cell.]

The key observations from the interference probability comparison are:

- 1) **Random baseline consistently fails**: With random frequency and power assignments, the interference probability remains near 1.0 (above 0.99) across all swarm sizes, confirming the necessity of intelligent spectrum planning even at small scales.
- 2) **Greedy methods show scale-dependent degradation**: The centralized greedy approach achieves excellent performance at $N = 10$ (interference probability ≈ 0.02) but degrades significantly as N increases, reaching approximately 0.27 at $N = 50$. This degradation stems from the sequential nature of greedy decisions: early choices cannot anticipate later conflicts, and the myopic local optimization leads to suboptimal global frequency reuse patterns. Greedy-Sequential performs similarly to Greedy-Central but slightly worse at larger scales. Greedy-Independent, lacking any neighbor awareness, performs substantially worse than the other greedy variants.
- 3) **IDDPG suffers from non-stationarity**: Independent DDPG training is unstable, with high variance across runs and frequent failure to converge. The lack of a centralized critic means each agent’s value function must implicitly model the changing policies of other agents, a task that becomes increasingly difficult as the number of agents grows. Credit assignment failures are common: an agent cannot distinguish between poor performance caused by its own suboptimal actions versus changes in other agents’ behaviors.

- 4) **CDDPG faces scalability limitations:** The centralized DDPG approach, which outputs $2N$ continuous actions from a single network, works reasonably for $N \leq 10$ but quickly becomes intractable. The joint action space dimensionality grows linearly with N , requiring exponentially more training data to cover the space adequately.
- 5) **MADDPG achieves superior scalability:** Our proposed MADDPG method [TO FILL: achieves interference probability of X at 10 UAVs, Y at 25 UAVs, Z at 50 UAVs], consistently outperforming all baselines at moderate to large swarm sizes. The CTDE architecture enables the critic to learn accurate value estimates using global information during training, while the decentralized actors scale independently of N at execution time.

C. Convergence Analysis

[Fig. 2: Training convergence curves. Top: Interference probability (y-axis) vs. training episodes (x-axis) for MADDPG at $N = 10, 25, 50$. Each curve shows the moving average of episode interference probability. Bottom: Evaluation interference probability at periodic evaluation checkpoints (every 25 episodes).]

The convergence analysis reveals several important patterns:

- **Temperature annealing drives convergence:** The initial warmup phase (temperature $\tau = 1.5$) produces highly stochastic behavior with interference probability near 1.0. As the temperature decays, the policy transitions from exploration to exploitation, and interference probability decreases steadily. The exponential decay schedule provides sufficient exploration time while eventually converging to deterministic behavior.
- **Larger swarms require more training:** At $N = 10$, the policy converges within approximately 150–200 episodes. At $N = 50$, convergence requires approximately 350–400 episodes, reflecting the increased complexity of coordinating more agents. We correspondingly scale the number of training episodes and warmup duration with swarm size.
- **Evaluation performance tracks training progress:** The periodic evaluation results show consistent improvement, indicating that the policy generalizes well to new random layouts rather than overfitting to training configurations. The evaluation interference probability decreases monotonically after the initial exploration phase.
- **IDDPG training instability:** In contrast, IDDPG training curves exhibit persistent oscillations and occasional divergence, with evaluation performance fluctuating significantly across checkpoints. This instability confirms the fundamental challenge of independent learning in non-stationary multi-agent environments.

D. Scalability Analysis

[Fig. 3: Scalability comparison. Left: Per-episode training wall-clock time (seconds) vs. UAV count for MADDPG,

IDDPG, CDDPG. Right: Peak memory usage (GB) vs. UAV count.]

The scalability analysis examines both computational and memory requirements:

- **Decentralized execution advantage:** During execution, each MADDPG agent uses only its local actor network with fixed observation dimension (42), independent of N . This means that once trained, the policy can be deployed on swarms of any size—new UAVs simply need their own instance of the same actor network. The computational cost per UAV at execution time is $O(1)$ with respect to N .
- **Training cost scales with N :** The centralized critic receives a global observation of dimension $42 \times N$ and joint actions of dimension $2N$, making training time scale approximately linearly with N . While this is significantly better than the combinatorial scaling of centralized optimization, it still imposes practical limits on training at very large scales ($N > 100$).
- **Memory efficiency:** The replay buffer stores transitions with global observations, requiring $O(N \cdot B \cdot C)$ memory where B is batch size and C is buffer capacity. For $N = 100$, this is approximately 5–8 GB, which is manageable on consumer hardware.

E. Ablation Studies

[Fig. 4: Ablation study results. Four sub-figures showing interference probability at $N = 25$ for: (a) Reward component ablation (full reward vs. global-only, vs. no-power-penalty, vs. no-congestion), (b) Number of observed neighbors K (3 vs. 5 vs. 10), (c) Observation radius R_{obs} (300 m vs. 600 m vs. 900 m), (d) Temperature schedule comparison (exponential decay vs. linear decay vs. constant).]

1) *Reward Design Ablation:* We ablate individual reward components by training MADDPG with simplified reward functions at $N = 25$:

- **Global only:** $r_i = -\lambda_g P_{\text{int}}$. Without per-agent terms, training converges more slowly and final interference probability is approximately [TO FILL] higher than the full reward, with some agents exhibiting free-riding behavior (high power, poor frequency choices).
- **No power penalty:** Removing r_i^{power} leads agents to use unnecessarily high transmit power, increasing global interference. Final interference probability increases by approximately [TO FILL].
- **No congestion term:** Removing $r_i^{\text{congestion}}$ degrades frequency diversity in dense regions, as agents lack explicit awareness of local interference conditions.
- **No high-SINR penalty:** Removing $r_i^{\text{high-SINR}}$ results in some agents transmitting at maximum power even when lower power would suffice, degrading global performance.

2) *Observation Design Ablation:* The number of observed neighbors K and observation radius R_{obs} control the trade-off between information richness and representation compactness:

- $K = 3$: With too few observed neighbors, agents lack information about relevant interferers that lie just outside the observation limiter, leading to suboptimal frequency choices.
- $K = 5$ (**default**): Provides a good balance, capturing the most significant interferers while maintaining compact observations.
- $K = 10$: Adding more neighbors provides diminishing returns while increasing observation dimensionality and training complexity.
- $R_{\text{obs}} = 300$ **m**: A shorter observation radius misses interferers at moderate distances whose cumulative effect is significant.
- $R_{\text{obs}} = 900$ **m**: A longer radius captures more information but adds marginal benefit since interference power decays with distance.

3) *Temperature Schedule Ablation*: Comparing exponential decay (our default) against linear decay and constant temperature reveals that exponential annealing provides the best balance of exploration and exploitation. Constant temperature ($\tau = 1.0$) never converges to deterministic behavior. Linear decay converges but requires more careful tuning of the decay rate.

F. Emergent Behavior Visualization

[Fig. 5: Emergent behavior visualization for $N = 50$ MADDPG policy. (a) Frequency heatmap: spatial map of the operational area with color indicating center frequency of each UAV. (b) Interference topology graph: nodes are UAVs, edges connect interferers to victims, edge width proportional to interference power. (c) SINR distribution histogram comparing random, greedy, and MADDPG policies. (d) Power distribution histogram for the same comparison.]

Analysis of the trained MADDPG policy reveals emergent coordination behaviors:

- **Spatial frequency reuse**: The frequency heatmap shows that UAVs in spatially separated regions naturally adopt similar frequencies, demonstrating that the policy learns to exploit spatial reuse. UAVs at close proximity tend to select well-separated frequencies.
- **Interference-aware power control**: The power distribution shows that MADDPG agents use lower average power than greedy methods while achieving better performance. This indicates that the policy learns to find frequency-separation solutions rather than compensating with high power.
- **Distributed channel allocation**: Without explicit channel assignment, the emergent behavior resembles a distributed graph coloring solution where the “colors” are frequency channels and the “conflict edges” are determined by mutual interference potential.
- **SINR distribution improvement**: The SINR histogram for MADDPG shows a tighter distribution centered above the threshold, with fewer outage events (SINR below threshold) compared to baseline methods.

V. DISCUSSION

A. Why MARL Works Better at Scale

The superior scalability of our MARL approach stems from three factors. First, the local observation design exploits the spatial locality of interference: a UAV’s transmission power decays with distance, so only nearby UAVs contribute significantly to interference. By restricting observations to the K nearest neighbors, we capture the most relevant coupling while avoiding the curse of dimensionality. Second, the CTDE architecture separates the learning of inter-agent dependencies (centralized critic) from the execution policy (decentralized actor), enabling the critic to reason about global interference patterns during training while the actor operates with minimal information at test time. Third, the Gumbel-Softmax parameterization aligns the action space with the discrete nature of spectrum allocation, avoiding the artifacts of continuous-to-discrete post-processing.

At small swarm sizes ($N \leq 10$), centralized greedy methods perform competitively because the number of conflict combinations is manageable. However, as N grows, the combinatorial nature of the joint optimization problem causes greedy methods to degrade, while the MARL policy—trained to handle diverse configurations—generalizes effectively.

B. Limitations

Several limitations of the current work warrant discussion. First, the training computational cost, while scalable, is still substantial for very large swarms ($N > 100$). The centralized critic’s input dimension grows linearly with N , and training would benefit from attention mechanisms or graph neural networks that can process variable-size global observations [30]. Second, the current approach assumes static UAV positions during each episode. In realistic scenarios, UAVs are mobile, and the frequency planning must adapt to changing topologies. Third, we have not yet compared against other MARL algorithms such as QMIX [24] or MAPPO [26], which employ value decomposition and centralized training with policy gradient, respectively, and may offer alternative trade-offs between coordination capability and training stability. Fourth, the method is sensitive to hyperparameters, particularly the temperature schedule and reward weights. Automated hyperparameter optimization would improve robustness to different deployment scenarios.

C. Practical Deployment Considerations

For practical deployment, several adaptations would be necessary. Online fine-tuning could enable the policy to adapt to specific deployment environments. Transfer learning from simulated to real environments would bridge the sim-to-real gap in radio propagation modeling. The sequential decision protocol naturally supports incremental swarm deployment, where new UAVs join an already-operating swarm and make decisions based on observations of existing members. Finally, integration with trajectory planning would create a joint optimization framework where UAV positions and spectrum assignments are co-optimized for mission objectives.

VI. RELATED WORK

A. Traditional Spectrum Allocation

Spectrum allocation has been extensively studied in the context of cellular networks and cognitive radio. Graph coloring approaches model interference as a binary conflict graph and assign orthogonal resources to conflicting links [6], [7]. While computationally efficient, these methods are inherently conservative because they treat any potential interference as a hard constraint rather than a soft optimization objective. Iterative water-filling [10] and game-theoretic approaches [11] enable distributed optimization but require explicit utility function definitions and may converge to inefficient Nash equilibria. Mixed-integer programming formulations [8] provide optimality guarantees but are computationally prohibitive for real-time application at scale.

B. Reinforcement Learning for Wireless Communications

Deep reinforcement learning has been increasingly applied to wireless communication problems. Single-agent DRL approaches have demonstrated success in dynamic spectrum access [14], [15], power control [16], and resource allocation [17]. However, these methods treat the environment as stationary from the agent's perspective, which breaks down in multi-transmitter scenarios where agents' actions are coupled through interference.

C. Multi-Agent Reinforcement Learning in Wireless Networks

MARL addresses the multi-agent nature of wireless resource allocation. The MADDPG algorithm [18] introduced the CTDE paradigm that has become standard in cooperative MARL. Subsequent work has applied MADDPG variants to power control in device-to-device (D2D) networks [19], UAV trajectory optimization [20], and spectrum sharing in cognitive radio networks [21]. Value decomposition methods such as QMIX [24] and VDN [25] factor the joint value function into per-agent utilities, enabling more efficient credit assignment in cooperative settings. MAPPO [26] extends proximal policy optimization to multi-agent settings with CTDE, achieving strong performance on cooperative benchmarks.

D. Large-Scale MARL Scalability

Scalability remains a fundamental challenge in MARL. As the number of agents grows, the joint observation-action space expands combinatorially. Recent work has explored mean-field approximations [29], where each agent interacts with the average effect of all other agents rather than modeling them individually. Attention mechanisms [30] enable agents to selectively focus on relevant neighbors, reducing the effective input dimensionality. Graph neural network-based policies [31] exploit the natural graph structure of multi-agent interactions and can generalize to varying numbers of agents.

E. UAV Communication and Swarm Coordination

UAV communication research has traditionally focused on trajectory optimization for communication performance [3], [4], air-to-ground channel modeling, and cellular-connected

UAVs [32]. Spectrum sharing among UAV swarms introduces additional challenges due to the three-dimensional mobility, dynamic topologies, and limited on-board processing. Recent work has investigated distributed spectrum sensing and access for UAV networks [33], but the joint optimization of frequency and power at scale remains an open problem. Our work contributes to this literature by providing a scalable MARL framework specifically designed for the large-scale UAV swarm frequency planning problem.

VII. CONCLUSION

In this paper, we addressed the challenge of interference-aware frequency planning in large-scale UAV swarms through a multi-agent reinforcement learning approach. We formulated the problem as a multi-agent MDP with local observations and proposed a MADDPG-based framework featuring Gumbel-Softmax discrete action parameterization, a sequential decision protocol, and a hybrid reward design. The CTDE architecture enables the centralized critic to learn inter-agent interference dependencies during training while allowing decentralized execution with only local observations, achieving scalability that centralized methods cannot match.

Experimental results across swarm sizes from 10 to 100 UAVs demonstrate that our approach [TO FILL: key quantitative results], consistently outperforming greedy baselines, independent RL, and centralized RL methods. Ablation studies confirm the importance of each design component: the hybrid reward structure, the local observation design with $K = 5$ neighbors within a 600 m radius, and the exponential temperature annealing schedule all contribute to the method's effectiveness. Visualization of learned policies reveals emergent spatial frequency reuse patterns that resemble distributed graph coloring solutions, indicating that the MARL agents learn meaningful coordination strategies.

Future work will extend this framework in several directions. First, we plan to incorporate UAV mobility by training policies on dynamic topologies and evaluating adaptive frequency reallocation. Second, we will investigate attention-based critic architectures to improve scalability to very large swarms ($N > 100$). Third, comparison with QMIX and MAPPO algorithms will provide a more complete picture of MARL method trade-offs in this domain. Finally, joint optimization of trajectory planning and spectrum allocation represents a natural extension that could further improve overall swarm communication performance.

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